Electronic Supplementary Material for "A Component-Level Audit of No-Reference Quality Scoring in a Multi-Video Summarization Pipeline"

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This supplement holds the detailed breakdowns behind results reported in the article, relocated here to keep the article at a readable length. Nothing in it is new: every figure quoted in the article is reproduced in full below, and the article cites these tables as Table S1, Table S2, and so on.

Table S1 In-domain collection: topics of Brazilian television news, with the number of source videos, segment slots, redundancy clusters and segment duration range. *Slots* counts cluster memberships; because two segments belong to two clusters each, the 33 slots correspond to 31 distinct segments.

Topic	Videos	Slots	Clusters	Duration (s)	Status
bebe_real	3	5	2	8–41	Included
brumadinho	3	6	3	13–188	Included
fone_canc._ruido	3	6	3	16–30	Included
isencao_imposto	3	6	3	25–30	Included
mounjaro	3	4	1	8–25	Included
ucrania	3	6	3	25–29	Included
esquema_pcc	3	–	–	–	Excluded[†]
Total	**18**	**33**	**15**	8–188	—

[†]98 SIFT descriptors $< K = 300$ K-means centroids. Slots exceed distinct segments in brumadinho and isencao_imposto, where one segment belongs to two clusters.

Table S1 (continued). Operational 42-cluster set used in the primary application analysis.

Topic	Clusters	Slots	Distinct segments
acordo_mariana	1	2	2
aumento_cafe	1	4	4
bebe_real	2	5	5
bolsonaro_trump	3	6	6
brasil_omc	1	2	2
brumadinho	3	6	5
catedral_notre_dame	2	5	4
cessar_fogo_ucrania	3	6	4
esquema_pcc	3	6	6
fone_cancelamento_ruido	3	6	6
fraude_inss	3	6	4
google_huawei	3	6	4
isencao_imposto	3	6	5
julgamento_bolsonaro	3	6	5
mounjaro	1	4	4
recorde_calor_2024	2	5	5
rio_celular	1	2	2
ucrania	3	6	6
ultraprocessados	1	4	4
Total	**42**	**93**	**83**

Table S2 Illustration of the sensitivity axis on a single broadcast frame and four degradations, after applying each method's declared polarity. The pattern this small case shows is the one Table S3 then measures over 25 distortion types and 10,125 images; nothing is concluded from this table alone. Cells are shaded by outcome: expected order, partial, violated.

Method	Pol.	Original	Blur	Noise	Dark	JPEG	Outcome
QualiCLIP	↑	0.341	0.155	0.167	0.196	0.163	as expected
LIQE	↑	1.294	1.056	1.000	1.023	1.023	as expected
BRISQUE	↓	42.8	90.0	66.4	59.1	78.2	as expected
NIQE	↓	4.53	11.99	15.79	6.18	9.47	as expected
Keypoint density	↑	2598	104	**4543**	4	2799	partial
DINOv2 ViT-S/14	↑	14.69	14.86	14.86	14.90	15.14	violated
ResNet-50	↑	15.62	**18.11**	15.00	15.97	15.28	violated

Values in bold are the ones that violate the expected order.

Table S2 (continued). Exact parameters of the constructed broadcast degradations.

Type	Levels 1–5
Gaussian blur	Kernel width $\{3, 5, 9, 15, 23\}$ pixels
White noise	Gaussian $\sigma = \{5, 12, 22, 35, 50\}$; deterministic seeds 1001–1005
Darkening	RGB multiplier $\{.80, .62, .46, .32, .20\}$ with clipping
JPEG	OpenCV quality $\{40, 25, 15, 9, 5\}$
H.264	libx264 CRF $\{30, 36, 41, 45, 50\}$, preset `veryfast`
Rescale	Downscale factors $\{.75, .55, .40, .28, .18\}$, then restore original size
Motion blur	Horizontal kernel width $\{5, 9, 15, 23, 33\}$ pixels
Desaturation	HSV saturation multiplier $\{.75, .55, .35, .18, .05\}$

Each segment contributes at most 30 uniformly spaced frames. The seeded 20-segment simple random sample represents 13 topics: one segment each from `aumento_cafe`, `bolsonaro_trump`, `cessar_fogo_ucrania`, `esquema_pcc`, `isencao_imposto`, `mounjaro` and `rio_celular`; two each from `bebe_real`, `google_huawei`, `julgamento_bolsonaro`, `recorde_calor_2024` and `ucrania`; and three from `fone_cancelamento_ruido`.

Table S3 Sensitivity to distortion level on KADID-10k. Direction-aligned SRCC is computed within each of the 81 reference images across five levels and then averaged by distortion type and across the 25 types. Intervals block-bootstrap the 81 references (10^4 resamples, seed 42). The last column counts types whose reference-averaged SRCC exceeds 0.50.

Method	Mean ρ^*	95% block CI	Types > 0.5
Q-Align	**0.880**	[0.874, 0.886]	**23/25**
QualiCLIP	**0.849**	[0.840, 0.857]	**22/25**
CLIP-IQA+	**0.808**	[0.795, 0.821]	**22/25**
LIQE	**0.807**	[0.797, 0.817]	**22/25**
MUSIQ	**0.733**	[0.720, 0.747]	**20/25**
TOPIQ	**0.721**	[0.702, 0.737]	**20/25**
HyperIQA	0.673	[0.653, 0.692]	**19/25**
DBCNN	0.668	[0.642, 0.692]	**19/25**
ARNIQA†	0.668	[0.652, 0.683]	**20/25**
BRISQUE	0.657	[0.628, 0.684]	**19/25**
CLIP-IQA	0.645	[0.612, 0.676]	**20/25**
TReS†	0.620	[0.598, 0.640]	**18/25**
NIQE	0.607	[0.587, 0.625]	**19/25**
SI	0.251	[0.219, 0.282]	13/25
Keypoint density	0.079	[0.033, 0.128]	7/25
SIFT+BoVW	−0.077	[−0.124, −0.030]	5/25
DINOv2	−0.064	[−0.136, 0.009]	1/25
ResNet-50	−0.297	[−0.364, −0.225]	0/25

†ARNIQA and TReS are reported on this axis as coverage extensions but are not treated as complete-case methods in the screening rule because the required CLIVE evaluations were not available.

Table S4 P.1401 point report after the logistic mapping. PLCC measures linearity, RMSE accuracy and the outlier ratio (OR; lower is better) consistency; RMSE* is the companion error after discounting the 95% confidence interval of the subjective score. OR saturates on KonIQ-10k because its per-image subjective intervals are narrow, but remains discriminative on KADID-10k. Analytic 95% intervals for PLCC, RMSE, RMSE* and OR are included in the released machine-readable results. A dagger marks evaluation on training data.

Method	KonIQ-10k				KADID-10k			
	PLCC	RMSE	RMSE*	OR	PLCC	RMSE	RMSE*	OR
Q-Align	.940	5.251	5.166	.983	.952	.330	.155	.292
LIQE	.915	6.234	6.149	.983	.720	.751	.526	.703
QualiCLIP	.771	9.837	9.752	.990	.673	.800	.572	.725
TOPIQ†	.978	3.198	3.124	.959	.546	.907	.671	.769
HyperIQA†	.956	4.507	4.428	.977	.503	.936	.700	.776
DBCNN†	.935	5.483	5.399	.982	.499	.939	.700	.799
MUSIQ†	.904	6.612	6.527	.985	.596	.869	.641	.730
CLIP-IQA+†	.863	7.790	7.705	.987	.665	.809	.581	.727
CLIP-IQA	.724	10.647	10.561	.992	.521	.924	.691	.787
SI	.561	12.782	12.696	.993	.328	1.023	.783	.821
SIFT+BoVW (canon.)	.484	13.508	13.423	.995	.224	1.055	.814	.835
Keypoint density	.484	13.502	13.417	.994	.227	1.055	.814	.836
NIQE	.393	14.194	14.108	.993	.439	.973	.734	.807
BRISQUE	.253	14.931	14.844	.995	.395	.995	.758	.807
ResNet-50	.162	15.231	15.143	.994	.194	1.062	.821	.834
DINOv2	.065	15.402	15.315	.996	.041	1.082	.841	.840

Table S5 Mean direction-aligned SRCC by paradigm. The two blocks are separated because the change between databases means different things in each. In the upper block no member was fitted on either database, so Δ compares two generalization results. In the lower block at least one member was fitted on KonIQ-10k, marked †; there the KonIQ-10k column mixes fit with generalization and no Δ is meaningful.

Paradigm	*N*	KADID	KonIQ	Δ	Best member
Neither database seen by any member					
ITU-T complexity	1	0.219	0.513	+0.294	SI
SIFT+BoVW*	2	0.124	0.446	+0.322	canonical IDF
Handcrafted	1	0.123	0.447	+0.324	Keypoint density
NR-IQA, NSS	2	0.355	0.302	−0.053	NIQE
CNN+BoVW	19	0.318	0.058	−0.260	VGG-19+BoVW
Deep features	2	−0.086	−0.115	−0.029	DINOv2 ViT-S/14
At least one member fitted on KonIQ-10k					
NR-IQA, LMM	1	**0.951**†	**0.920**†	—	Q-Align
NR-IQA, CNN reg.	2	0.473	**0.933**†	—	HyperIQA
NR-IQA, transformer	2	0.534	**0.923**†	—	TOPIQ
NR-IQA, CLIP	4	0.636	**0.774**†	—	LIQE

*Mean over the magnitudes of the two TF-IDF variants, which differ in orientation but not in strength. Restricted to the two CLIP members fitted on neither database, that paradigm gives 0.583 on KADID-10k and 0.679 on KonIQ-10k, a change of +0.096.

Table S6 The BoVW variants in the domain, on the agreement axis and on what they deliver. *Differs* counts the clusters in which the variant selects a different segment from the deployed one; *Agreement* is top-1 agreement with the reference model over the 15 clusters; r_{kp} is the rank correlation with keypoint density over the 33 segment slots; and the last three columns are the summary the variant delivers, as in Table S11. The comparison between summing and averaging defined in the main article is therefore evaluated directly. The final column is the uncalibrated raw sum of the DOVER technical and aesthetic branches, used only as a scale-sensitivity diagnostic.

Variant	Aggr.	Differs	Agreem.	r_{kp}	Dur. (s)	Q-Align	raw T+A
SIFT+BoVW (as implemented)	sum	—	9/15	+0.591	634.0	61.1%	52.1%
SIFT+BoVW (canonical IDF)	sum	1/15	10/15	+0.116	801.5	62.2%	47.6%
BoVW / descriptor count	mean	2/15	10/15	−0.199	810.9	59.0%	48.2%
BoVW / frame count	mean	1/15	10/15	−0.145	**459.3**	**77.3%**	**71.5%**
Keypoint density	mean	5/15	7/15	+1.000	—	—	—
LIQE (for reference)	mean	5/15	11/15	−0.381	455.6	85.8%	91.8%

Table S7 Descriptive in-domain agreement with the Q-Align reference model over the 15 human-annotated redundancy clusters, raw and chance-corrected against $\bar{p}_e = 0.472$. These clusters share topics and two pairs share a segment; the table therefore makes no cluster-independence inference. Topic-block intervals for the larger system-defined set are reported in the main article. Roles follow Table 1: D = deployed, C = candidate and X = diagnostic.

Method	Role	Agreement	$\kappa_{\mathrm{uniform}}$
MUSIQ	C	13/15 (86.7%)	+0.747
DBCNN	C	12/15 (80.0%)	+0.621
TOPIQ	C	12/15 (80.0%)	+0.621
LIQE	C	11/15 (73.3%)	+0.495
HyperIQA	C	11/15 (73.3%)	+0.495
NIQE	C	10/15 (66.7%)	+0.368
QualiCLIP	C	10/15 (66.7%)	+0.368
BRISQUE	C	10/15 (66.7%)	+0.368
SIFT+BoVW	D	9/15 (60.0%)	+0.242
CLIP-IQA	C	8/15 (53.3%)	+0.116
CLIP-IQA+	C	8/15 (53.3%)	+0.116
DINOv2	X	5/15 (33.3%)	−0.263

Table S8 The four redundancy clusters in which the method under study diverges from the majority of the quality references. Each row is a candidate segment; each column a scoring method. The top-1 choice of each method is shaded— pink for SIFT+BoVW, green for the quality references. NIQE and BRISQUE are lower-is-better. Scores are reproduced from the aggregated per-cluster results.

Segment	SIFT+BoVW	LIQE	NIQE↓	Q-Align	QualiCLIP	BRISQUE↓
bebe_real C1						
Band 0:37 (10 s)	**5448.0**	1.49	6.55	4.02	0.28	50.97
Record 0:46 (11 s)	2077.6	**1.59**	**5.57**	**4.17**	**0.30**	**41.44**
brumadinho C2						
Band 0:00 (13 s)	2888.3	**2.60**	**5.04**	**4.84**	**0.36**	**44.95**
Record 4:24 (188 s)	**5332.7**	1.13	6.23	3.74	0.27	53.57
isencao_imposto C0						
Band 2:46 (30 s)	**3732.8**	4.23	**4.67**	3.70	**0.62**	34.65
CNN 1:12 (25 s)	3010.0	**4.47**	4.74	**4.26**	0.59	**29.84**
mounjaro C0						
Band 0:00 (8 s)	559.8	4.12	7.29	**4.38**	0.56	63.19
Dom. Esp. 2:06 (16 s)	**1945.1**	3.95	5.87	3.90	0.57	36.55
Dom. Esp. 3:07 (25 s)	1784.4	3.61	5.28	3.68	0.56	42.29
Record 0:00 (14 s)	815.2	**4.93**	**4.76**	4.26	**0.63**	**35.45**

Table S9 Top-1 agreement with the reference model over the 15 clusters under four temporal poolings. The worst-frame, tenth-percentile and mean − sd rules all penalise within-segment dispersion; the mean ignores it.

Method	Mean	Worst frame	10th pct.	mean − sd
LIQE	11/15	11/15	11/15	11/15
NIQE	10/15	10/15	10/15	10/15
QualiCLIP	9/15	9/15	11/15	10/15
BRISQUE	10/15	9/15	9/15	10/15

Table S10 Eligible set as a function of the generalization threshold, under both readings of "a database it was not fitted on". *Universal* requires the threshold to hold on every unseen database (the reading adopted in the main article); *existential* requires it on at least one. The sensitivity requirement is held at a within-reference mean above 0.50 and at least 15/25 types; the topic-block lower confidence limit must also exceed the in-domain floor.

Minimum ρ^*	Universal reading (adopted)	Existential reading
0.50	LIQE, QualiCLIP, CLIP-IQA+, MUSIQ, TOPIQ, CLIP-IQA	LIQE, QualiCLIP, CLIP-IQA+, MUSIQ, TOPIQ, CLIP-IQA, DBCNN, HyperIQA
0.55	LIQE, QualiCLIP, CLIP-IQA+, MUSIQ	*same eight methods*
0.60	**LIQE, QualiCLIP, CLIP-IQA+**	*same eight methods*
0.65*	**LIQE, QualiCLIP, CLIP-IQA+**	*same eight methods*
0.70	LIQE	*same eight methods*
0.75	none	LIQE, QualiCLIP, CLIP-IQA+, MUSIQ, TOPIQ, DBCNN, HyperIQA

*The value used. At the adopted generalization threshold and universal reading, the same three methods survive for sensitivity type-count cuts from 12/25 through 22/25; none survives at 23/25. Requiring only a positive floor point estimate instead of a topic-block lower limit does not change the three-method set.

Table S11 The same comparison on the 15 human-annotated clusters. The two references agree closely here; their disagreement emerges on the 42-cluster analysis in the main article, which is the primary estimate. The DOVER column uses the uncalibrated raw technical-plus-aesthetic sum, not DOVER-T.

Scoring method	Changed	by raw T+A	by Q-Align	Dur. (s)	Words
SIFT+BoVW (deployed)	—	52.1%	61.1%	634.0	1,613
LIQE	5/15	91.8%	85.8%	455.6	1,124
BRISQUE	6/15	84.5%	84.7%	431.0	1,073
QualiCLIP	4/15	80.9%	80.4%	439.6	1,124
NIQE	4/15	80.9%	80.4%	439.6	1,124
DINOv2	12/15	53.4%	36.7%	328.6	904

Table S11 (continued). Construct and scale sensitivity on the 42 operational clusters.

DOVER output	Estimand	SIFT	LIQE	Δ; topic CI	Hierarchical CI
Technical branch	Global range (%)	50.5	76.0	+25.5; [−12.1, 54.9]	[−17.2, 59.6]
Technical branch	Raw mean	−.0770	−.0701	+.0069; [−.0028, .0178]	[−.0041, .0197]
Technical branch	Mean cluster position (%)	57.1	64.4	+7.2; [−20.7, 35.1]	[−22.4, 37.5]
Technical branch	Selected maximum (%)	54.8	59.5	+4.8; [−22.2, 31.7]	[−24.4, 34.2]
Raw T+A sum	Global range (%)	47.4	77.7	+30.3; [1.1, 54.1]	[−4.1, 59.0]
Aesthetic branch	Global range (%)	45.0	73.7	+28.7; [5.3, 49.7]	[−.2, 54.6]

The branch sum is the raw technical output plus the raw aesthetic output. It is not the official ITU-standardized DOVER fused score. The technical branch is primary; the other rows diagnose construct sensitivity.

DOVER-T range filter	Clusters	Threshold	Global-range Δ
All clusters	42	—	+25.5
Exclude bottom 10%	37	.00397	+25.9
Exclude bottom 25%	31	.00891	+28.1

DOVER-T cluster ranges: minimum .00028, first quartile .00891, median .01686, third quartile .04118, maximum .10653; zero-range clusters: 0.

Matched-input control: ten identical frames and mean pooling.

Scorer	Differs from SIFT	DOVER-T range	Raw mean	Selected max
SIFT+BoVW	—	52.0%	−.0766	57.1%
LIQE	16/42	79.1%	−.0692	64.3%

LIQE–SIFT global-range Δ = +27.1 points (topic CI [5.6, 47.8]; hierarchical CI [2.4, 52.5]). Raw-branch Δ = +.00736 (topic [.00135, .01406]; hierarchical [.00057, .01683]). Thirteen of the 16 disagreement clusters belong to the 22-cluster complete-operation change set. Relative to the published deployed path, matched-input SIFT changes 12 choices and matched-input LIQE changes 22; the control isolates scorer identity, not the historical operation itself.

Table S12 The eight predictions registered at commit `9377c07` before the constructed experiment, against the measurements. None is used to filter a method.

	Subject	Prediction	Measured	Outcome
P1	SIFT+BoVW	High accuracy on blur and darkening	90%, 100%	holds
P2	SIFT+BoVW	Inverted on white noise	8%	holds
P3	SIFT+BoVW	Near chance on JPEG and H.264	6% and 90%	refuted
P4	CNN+BoVW (19)	Chance on every type	44.9%, CI [41.0, 48.9]; range 30–64	refuted
P5	NIQE, BRISQUE	Accuracy falls under heavy overlay	+2.0, +2.2 points	refuted
P6	Learned methods	No interaction with overlay	−18.1 to −8.1 for three	refuted
P7	SIFT+BoVW	Design B mean below 0.15	−0.425	refuted
P8	Q-Align	Design A accuracy above 0.90	98.2%, CI [97.2, 99.1]	holds

The complete registered commit is:

`9377c07ca080ac61c7cff80e0ff6c7507b6ebaba`.

The submission archive contains `registered_prediction_commit.bundle`; the clone, bundle verification and object-inspection commands in its README make the commit and timestamp independently inspectable even before the public release.

Table S12 (continued). Fidelity and descriptor-count diagnostics for the constructed broadcast pairs.

Method	All 8 types (800)			Without dark/desaturation (600)		
	Correct	Ties	Accuracy	Correct	Ties	Accuracy
SIFT+BoVW	521	0	65.1%	382	0	63.7%
LIQE	796	0	99.5%	599	0	99.8%
Q-Align	786	0	98.2%	600	0	100.0%
DOVER-T	781	0	97.6%	599	0	99.8%

Type	Mean ΔD	$\Delta D > 0$	Mean ΔD by level 1–5
Blur	−797	10%	−130, −417, −913, −1188, −1339
Dark	−796	0%	−229, −487, −791, −1116, −1358
H.264	−398	10%	−128, −220, −361, −541, −737
JPEG	+289	95%	+161, +252, +339, +376, +317
Motion blur	−1067	0%	−548, −909, −1174, −1308, −1396
Noise	+469	92%	+126, +348, +517, +607, +749
Rescale	−646	12%	−154, −377, −585, −911, −1204
Desaturation	+18	61%	0, +14, +21, +34, +24

$\Delta D = D_{\text{degraded}} - D_{\text{unmodified}}$. Strict accuracy uses correct/total; a tie is neither correct nor incorrect and remains in the denominator.

DOVER-T	dark	motion blur	H.264	rescale	satur.	noise	JPEG	All
Accuracy (%)	93	100	100	100	100	89	99	100	97.6
Mean severity ρ^*	.985	.900	1.000	1.000	1.000	.715	.995	1.000	—

The raw technical output is the exact branch later used for the 42-cluster output analysis. Its overall segment-block 95% CI is [96.1, 99.0]; after excluding darkening and desaturation, accuracy is 99.8% with CI [99.5, 100]. There are no ties. Each condition is encoded through the same 5-fps MP4 wrapper and uses paired spatial fragment seed 42.

Table S13 Method inventory and released-weight supervision. *Sup.*: OU = opinion-unaware, K = KonIQ-10k, L = LIVE, M = multi-database mixture. Roles follow Table 1: D = deployed, C = candidate, R = reference and X = diagnostic or ablation. The daggered partial-coverage methods are excluded from complete-case screening.

Method	Family	Construct	Sup.	Pol.	GPU	Role
SIFT+BoVW	Handcrafted	TF-IDF word count	–	↑	No	D
Keypoint density	Handcrafted	keypoints/frame	–	↑	No	X
DINOv2	Self-supervised	patch-token norm	–	↑	Yes	X
ResNet-50	Supervised	pooled-feature norm	–	↑	Yes	X
BRISQUE	NSS	natural-statistics deviation	L	↓	No	C
NIQE	NSS	pristine-model distance	OU	↓	No	C
QualiCLIP	CLIP	quality score	OU	↑	Yes	C
LIQE	CLIP	quality, distortion, scene	K	↑	Yes	C
Q-Align	LMM	textual quality levels	M	↑	Yes	R
SI	ITU-T complexity	Sobel energy	–	↑	No	X
19 CNN+BoVW	ablation	TF-IDF word count	–	↑	Yes	X
DBCNN	CNN regression	bilinear pooling	K	↑	Yes	C
HyperIQA	CNN regression	adaptive weights	K	↑	Yes	C
MUSIQ	Transformer	multi-scale score	K	↑	Yes	C
TOPIQ	Transformer	semantic attention	K	↑	Yes	C
CLIP-IQA	CLIP	antonym prompts	OU	↑	Yes	C
CLIP-IQA+	CLIP	tuned prompts	K	↑	Yes	C
ARNIQA†	Self-supervised	distortion manifold	K	↑	Yes	partial
TReS†	CNN–transformer	ranking/consistency	K	↑	Yes	partial

PyIQA registry keys used by the archived scorers are `brisque`, `niqe`, `dbcnn`, `hyperiqa`, `musiq`, `topiq_nr`, `clipiqa`, `clipiqa+`, `liqe`, `qualiclip`, `arniqa` and `tres`. The final inventory and analysis archive were frozen on 23 August 2026. MANIQA was attempted with PyIQA 0.1.15.post2 and the `maniqa-kadid` checkpoint. A single-image diagnostic completed, but the released implementation evaluates 20 crops per image and the full 21,360-image three-database run did not finish within the experiment window on the single 8-GB RTX 4060; it is therefore an uncompleted method, not a failed candidate.

Table S13 (continued). Complete extractor inventory for the CNN+BoVW ablation. Every row uses the final spatial tensor before global pooling or classification.

Extractor	Released torchvision weights	Spatial endpoint
ResNet-18	IMAGENET1K_V1	trunk before average pooling
ResNet-50	IMAGENET1K_V2	trunk before average pooling
ResNet-101	IMAGENET1K_V2	trunk before average pooling
VGG-16	IMAGENET1K_V1	full `features` module
VGG-19	IMAGENET1K_V1	full `features` module
EfficientNet-B0	IMAGENET1K_V1	full `features` module
MobileNet-V3-Large	IMAGENET1K_V2	full `features` module
ConvNeXt-Tiny	IMAGENET1K_V1	full `features` module
ConvNeXt-Small	IMAGENET1K_V1	full `features` module
DenseNet-121	IMAGENET1K_V1	full `features` module
DenseNet-201	IMAGENET1K_V1	full `features` module
ResNeXt-50-32x4d	IMAGENET1K_V2	trunk before average pooling
ResNeXt-101-64x4d	IMAGENET1K_V1	trunk before average pooling
Wide-ResNet-50-2	IMAGENET1K_V2	trunk before average pooling
Wide-ResNet-101-2	IMAGENET1K_V2	trunk before average pooling
Inception-V3	IMAGENET1K_V1	convolutional trunk through Mixed_7c
EfficientNet-B4	IMAGENET1K_V1	full `features` module
SqueezeNet-1.1	IMAGENET1K_V1	full `features` module
MobileNet-V2	IMAGENET1K_V2	full `features` module

Common preprocessing: bicubic resize to 256 pixels, centre crop to 224×224, ImageNet mean and standard deviation, and FP16 inference. Each spatial location is flattened into one descriptor and ℓ_2-normalised in FP32. The same internal key-frame filter, vocabulary construction, TF–IDF weighting and segment aggregation are then applied. The archived implementation is `cnn_bovw_scorer.py`.

Table S13 (continued). Exact configurations of the two model-based references used in the broadcast analyses.

Model	Config.	Input / pooling	Output used
Q-Align	`q-future/one-align`; task `quality`, input `image`	Uniform 1-fps frames; arithmetic mean of frame scores	Four-bit NF4 with double quantisation and FP16 compute; higher score is better. The same configuration scores the 800 constructed pairs and supplies the in-domain floor.
DOVER	Official `DOVER.pth` and `dover.yml`, `val-l1080p` view decomposition	Technical: 7×7 spatial fragments of size 32, 32 frames, interval 2, three clips; aesthetic: 224×224, 32 frames, interval 2, one clip	The technical branch is the primary output reference and is also evaluated on the 800 pairs with a common 5-fps MP4 wrapper and paired seed 42. The raw branch sum is only a sensitivity analysis; it is not the official standardised fusion or DOVER++.

Table S14 Computational cost measured on one RTX 4060 over the same inputs. Frame-wise entries report median latency after ten warm-up and sixty timed iterations; the final two rows are measured complete paths.

Method	ms/frame	IQR	Per cluster
SI	2.2	0.14	0.2 s
QualiCLIP	14.3	1.68	1.2 s
TOPIQ	18.7	0.36	1.5 s
BRISQUE	19.6	0.98	1.6 s
CLIP-IQA+	21.4	0.65	1.8 s
DBCNN	21.5	0.46	1.8 s
LIQE	22.7	0.82	1.9 s
CLIP-IQA	25.2	0.51	2.1 s
NIQE	34.6	0.90	2.8 s
MUSIQ	40.0	9.36	3.3 s
ResNet-50	46.2	8.44	3.8 s
DINOv2	46.4	36.63	3.8 s
Keypoint density	46.8	1.48	3.8 s
HyperIQA	50.1	0.77	4.1 s
SIFT+BoVW	$\sim$ 1,724	—	182 s
Q-Align	322.6	—	26 s

The uniform 1-fps candidate cache contains 2,892 frames from 83 distinct segments. Counting repeated memberships across the 93 cluster slots gives 3,525 segment-frame instances. Candidate latency was timed per decoded frame; the SIFT+BoVW row is the complete operation divided by its input-frame count and therefore includes key-frame selection, descriptor extraction, vocabulary quantisation and aggregation. The per-cluster column is the operational quantity; the per-frame SIFT value is an approximate normalisation rather than an identical-path microbenchmark.

Table S15 Descriptive three-database contrasts and released-weight exposure. K–C is KonIQ-10k minus CLIVE; C–KADID is CLIVE minus KADID-10k. Positive values are losses. K = KonIQ-10k opinion training, OU = opinion-unaware and L = LIVE opinion training. The contrasts do not isolate causal effects because the databases, architectures and objectives are unmatched.

Method	Sup.	KonIQ	CLIVE	KADID	K–C	C–KADID
Released checkpoint fitted on KonIQ-10k						
TOPIQ	K	0.973†	0.811	0.511	+0.162	+0.300
HyperIQA	K	0.947†	0.750	0.462	+0.197	+0.288
DBCNN	K	0.919†	0.756	0.484	+0.163	+0.272
MUSIQ	K	0.873†	0.789	0.556	+0.084	+0.233
CLIP-IQA+	K	0.824†	0.805	0.654	+0.019	+0.151
LIQE	K	0.915†	0.865	0.723	+0.050	+0.142
mean over the six					+0.113	+0.231
Never fitted on KonIQ-10k						
QualiCLIP	OU	0.711	0.757	0.665	−0.046	+0.092
CLIP-IQA	OU	0.647	0.703	0.501	−0.056	+0.202
NIQE	OU	0.368	0.449	0.379	−0.081	+0.070
BRISQUE	L	0.236	0.313	0.330	−0.077	−0.017
mean over the four					−0.065	+0.087

†In-distribution. CLIVE contains 1,162 images and is unseen by every checkpoint in the table. The operational partition assigns 33% of the mean KonIQ-to-KADID gap of the fitted group to K–C and 67% to C–KADID; these percentages describe contrasts and are not causal estimates.

Table S16 Spearman association with human scores before and after conditioning on detectable keypoint density. r_{kp} is the rank correlation between the method score and keypoint density. Differences are descriptive attenuations, not causal fractions of "content density."

Method	KonIQ-10k				KADID-10k			
	r_{kp}	$\lvert\rho\rvert$	$\lvert\rho_{\cdot\mathrm{kp}}\rvert$	Diff.	r_{kp}	$\lvert\rho\rvert$	$\lvert\rho_{\cdot\mathrm{kp}}\rvert$	Diff.
Q-Align	+.44	.920	.901	−.019	+.13	.951	.950	−.001
LIQE	+.43	.915	.894	−.020	+.18	.723	.718	−.005
TOPIQ	+.48	.973	.966	−.007	+.21	.511	.500	−.011
HyperIQA	+.48	.947	.933	−.014	+.17	.462	.451	−.011
DBCNN	+.49	.919	.898	−.021	+.22	.484	.472	−.012
MUSIQ	+.44	.873	.841	−.032	+.20	.556	.546	−.010
QualiCLIP	+.53	.711	.626	−.086	+.29	.665	.663	−.002
CLIP-IQA	+.27	.647	.611	−.036	+.22	.501	.489	−.012
SI	+.77	.513	.296	−.218	+.70	.219	.187	−.032
NIQE	−.63	.368	.125	−.243	−.33	.379	.362	−.018
BRISQUE	−.31	.236	.116	−.120	−.13	.330	.319	−.011
SIFT+BoVW (canonical)	+.998	.447	.016	arith.	+.992	.125	.028	arith.
SIFT+BoVW (deployed)	−.9999	.445	n.d.	arith.	−.9998	.123	n.d.	arith.

The shaded rows are the near-tautological regime: conditioning a score on a variable with which it is almost perfectly correlated removes the variable from itself. For the deployed row the partial coefficient is not reported because the denominator approaches zero and the result is numerically unstable.